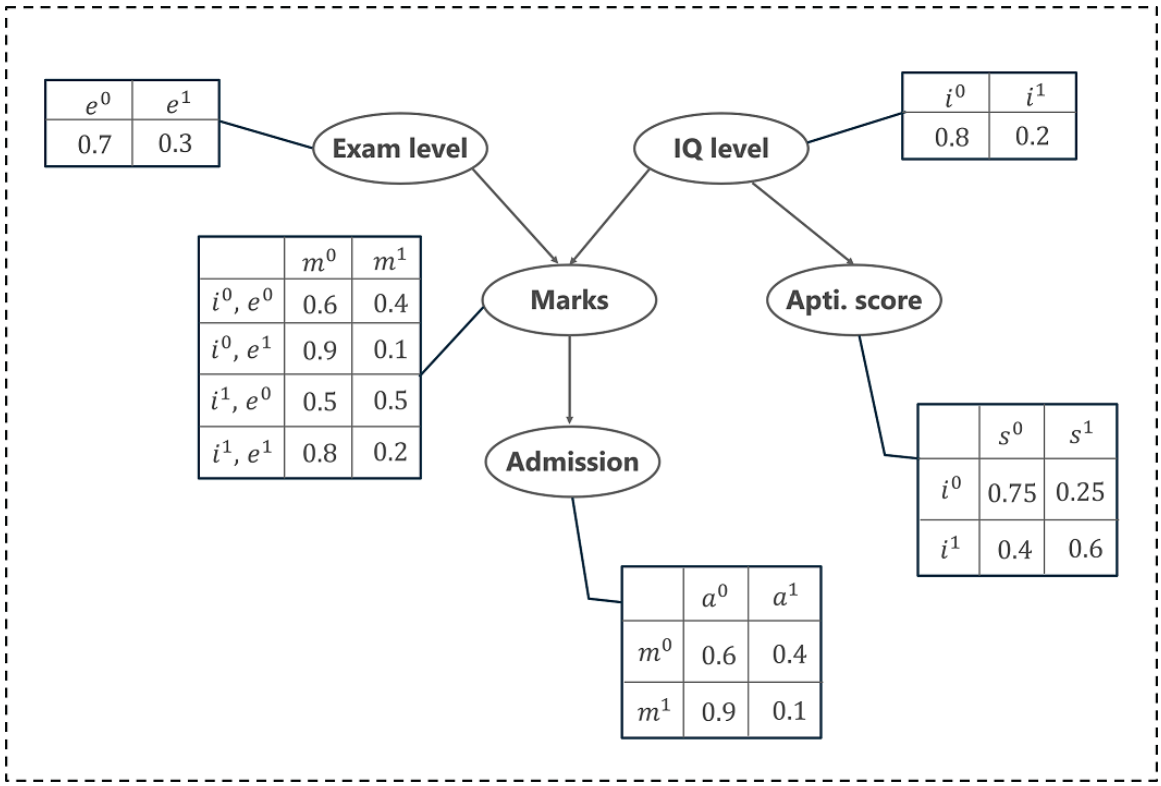
Problem Set 3 - Uncertainty

Introduction to Artificial Intelligence
CSAC 212

1 Bayesian Network

Implement the given Bayesian network below in Python code using *pgmpy* library. Refer to the code given during the lecture session or refer to the documentation on this link: https://pgmpy.org/models/bayesiannetwork.html.



e^0	e^1
0.7	0.3

i^0	i^1
0.8	0.2

	m^0	m^1
i^0, e^0	0.6	0.4
i^0, e^1	0.9	0.1
i^1, e^0	0.5	0.5
i^1, e^1	0.8	0.2

	s^0	s^1
i^0	0.75	0.25
i^1	0.4	0.6

	a^0	a^1
m^0	0.6	0.4
m^1	0.9	0.1

2 Problem

Answer the given problems below. Use markdown to explain your code and analysis. Submit your code as PS3_LastName_FirstName.ipynb

1.1 (5 points) Find the independencies and the active trail nodes of the bayesian network.

1.2 (20 points) Explore variable elimination by using different queries and evidences. This is in free form. Analyze and justify the explored parameters.

1.3 (5 points) Get the most probable state of all the variables.